

Spring 2026. Physics 510.

Quantum Theory II.

Homework # 4.

Due to March 25th, 2026.

1. **Born approximation. Desai book, Chapter 20**, problems 20.1-20.4.
2. **Born approximation and Pauli Principle. Desai book, Chapter 20**, problem 20.5. Hint: Recall the (anti)symmetrization requirement for a wave function and its spin and coordinate factors. Consider both singlet and triplet states.
3. **Partial phase shifts. Desai book, Chapter 21**, problems 21.1, 21.2, 21.4.

Do not copy the answers from the books. Do all calculations till tiny details.